

● HARD

● ADVANCED MATH

Nonlinear equations in one variable and systems of equations in two variables

03

10 questions · ~15 min

Q1

ADVANCED MATH

$$y = x + 9$$

$$y = x^2 + 16x + 63$$

A solution to the given system of equations is x, y . What is the greatest possible value of x ?

- A. -6
- B. 7
- C. 9
- D. 63

Q2

GRID-IN ADVANCED MATH

$$x^2 + y + 7 = 7$$

$$20x + 100 - y = 0$$

The solution to the given system of equations is x, y . What is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$D = T - \frac{9}{25}(100 - H)$$

The formula above can be used to approximate the dew point D , in degrees Fahrenheit, given the temperature T , in degrees Fahrenheit, and the relative humidity of H percent, where $H > 50$. Which of the following expresses the relative humidity in terms of the temperature and the dew point?

A. $H = \frac{25}{9}(D - T) + 100$

B. $H = \frac{25}{9}(D - T) - 100$

C. $H = \frac{25}{9}(D + T) + 100$

D. $H = \frac{25}{9}(D + T) - 100$

Q4

GRID-IN

ADVANCED MATH

$$64x^2 - 16a + 4bx + ab = 0$$

In the given equation, a and b are positive constants. The sum of the solutions to the given equation is $k4a + b$, where k is a constant. What is the value of k ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

ADVANCED MATH

$$64x^2 + bx + 25 = 0$$

In the given equation, b is a constant. For which of the following values of b will the equation have more than one real solution?

- A. -91
- B. -80
- C. 5
- D. 40

Q6

ADVANCED MATH

$$\sqrt{2x+6} + 4 = x + 3$$

What is the solution set of the equation above?

- A. $\{-1\}$
- B. $\{5\}$
- C. $\{-1, 5\}$
- D. $\{0, -1, 5\}$

Q7

GRID-IN

ADVANCED MATH

$$y = -2.5$$

$$y = x^2 + 8x + k$$

In the given system of equations, k is a positive integer constant. The system has no real solutions. What is the least possible value of k ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$2x^2 - 2 = 2x + 3$$

Which of the following is a solution to the equation above?

A. 2

B. $1 - \sqrt{11}$

C. $\frac{1}{2} + \sqrt{11}$

D. $\frac{1 + \sqrt{11}}{2}$

Q9

GRID-IN ADVANCED MATH

$$5(x + 7) = 15x - 17x + 7$$

What is the sum of the solutions to the given equation?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$x - 1^2 = -4$$

How many distinct real solutions does the given equation have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero